Semester	B.E. Semester VII – Computer Engineering
Subject	DevOps Lab
Subject In-charge	Dr. Snehal Andhare

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Grade and Subject Teacher's Signature	

Task Number	9
Project Title:	Secure CI/CD Pipeline for a Loan Application Tracking Portal
Task Title	Selenium Test Design and Local Execution: Identify three to five critical user journeys in the Loan Application Tracking Portal. Write Selenium WebDriver test cases with assertions, test data and screenshots for failures. Execute the suite locally through Maven.
Deliverable:	

Selenium Test Design and Local Execution

1. Selenium Test Plan:

Journey #	Test Journey Name	Target Route	Actions & Inputs	Assertions & Quality Gate
UJ-01	**Application Intake & Event Registration**	`/apply`	Fill Name (*"Rahul Sharma"*), Email, Phone, Amount (*₹45,000*), Category (*"PERSONAL"*), Click Submit.	Verify redirect to `/dashboard`; assert new row displays generated LOAN-XXXX ID and status `SUBMITTED`.
UJ-02	**Search & Real-time Filter**	`/dashboard`	Type *"Rahul"* into `searchBox`, click `searchBtn`.	Assert table contains 1 matching row with *"Rahul Sharma"*; assert non-matching records are excluded.
UJ-03	**Status Drill-Down & Timeline Verification**	`/loans/{id}`	Click *Drill-Down / Timeline →|rightarrow→*, select *"DOCUMENT_VERIFIED"*, enter remarks, click Update.	Assert status badge changes to `DOCUMENT_VERIFIED`; assert audit timeline count

				increases by 1 with timestamp.
UJ-04	**Rejection & Exception Alerts Queue**	`/loans/{id}` →|*rightarrow*→ `/alerts`	Update status to *"REJECTED"* with reason *"CIBIL Score Below 650"*, navigate to `/alerts`.	Assert `/alerts` displays the loan row with red badge containing reason *"CIBIL Score Below 650"*.

2. Creating the Test Components

Created Selenium Testing Components

1. `LoanPortalSeleniumE2ETest.java`
   - Runs in **Headless Chrome** (zero browser popups, ideal for automated CI servers).
   - Executes all **4 critical user journeys** in sequence:
     - **Journey 1:** Form intake on `/apply` → verifies dashboard listing and generated `LOAN-XXXX` ID.
     - **Journey 2:** Real-time search query on `/dashboard` → verifies filtering and reset.
     - **Journey 3:** Status drill-down on `/loans/{id}` → advances stage to `DOCUMENT_VERIFIED` and verifies the event timeline.
     - **Journey 4:** Credit rejection → routes application into the `/alerts` exception view with rejection reason.
2. `ScreenshotOnFailureExtension.java`
   - JUnit 5 `TestWatcher` callback that automatically snaps full-page PNG screenshots and saves them to `target/screenshots/` if any assertion fails.

Step-by-Step Local Execution:

1: Execute the Selenium Suite Locally

Inside CMD : mvn test -Dtest=LoanPortalSeleniumE2ETest

```
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 12.97 s -- in com.loanportal.selenium.LoanPortalSeleniumE2ETest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  16.583 s
[INFO] Finished at: 2026-10-15T01:22:33+05:30
[INFO] ------------------------------------------------------------------------
```

2: Run All Tests (Unit + Selenium)

mvn clean test

```
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.139 s -- in com.loanportal.service.LoanServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 9, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  19.050 s
[INFO] Finished at: 2026-10-15T01:24:02+05:30
[INFO] ------------------------------------------------------------------------
```

3: Commit and Push the Selenium Suite to GitHub

```
D:\Vicky_Project\DevOps\DevOps_Project>git add src/test/java/com/loanportal/selenium/
warning: in the working copy of 'src/test/java/com/loanportal/selenium/LoanPortalSeleniumE2ETest.java', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of 'src/test/java/com/loanportal/selenium/ScreenshotOnFailureExtension.java', LF will be replaced by CRLF the next time Git touches it

D:\Vicky_Project\DevOps\DevOps_Project>git commit -m "test(selenium): add headless E2E test suite for 4 critical user journeys with screenshot capture"
[main 9b103a5] test(selenium): add headless E2E test suite for 4 critical user journeys with screenshot capture
 2 files changed, 237 insertions(+)
 create mode 100644 src/test/java/com/loanportal/selenium/LoanPortalSeleniumE2ETest.java
 create mode 100644 src/test/java/com/loanportal/selenium/ScreenshotOnFailureExtension.java

D:\Vicky_Project\DevOps\DevOps_Project>git push origin main
Enumerating objects: 16, done.
Counting objects: 100% (16/16), done.
Delta compression using up to 16 threads
Compressing objects: 100% (7/7), done.
Writing objects: 100% (10/10), 3.48 KiB | 1.74 MiB/s, done.
Total 10 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/vickypukale/loan-application-tracking-portal.git
   15df393..9b103a5  main -> main
```